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Unlocking Hidden Sands: Seismic Interpretation Strategies for Thin Challenging Sand Deposit

Marwa Al Harthy, Mohammed Al Harrasi, Ahmed Al Naabi, Khamis Al Subhi, Samir Al Sulaimi, Khamis Al Wahaibi, Azzan Al Harrasi, Mohamed AbdelGhaffar, and Hilal Al Yaarubi, Daleel Petroleum LLC, Muscat, Sultanate of Oman

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Abstract

This paper presents seismic challenges in unlocking a previously unrecognized shallow thin sandstone reservoir in the Shammar Formation, deposited directly above base Cenozoic/Cretaceous unconformity. Although penetrated by wells targeting deeper Formations, it remained unidentified due to limited data and lack of geological focus. Dense seismic interpretation coupled with well data integration and drilling results, revealed thin, small sand basins in fault-bounded, leading to unlocking new opportunities and maturing additional hydrocarbon volumes.

Fine-tuned seismic interpretation of the unconformity carried out using different band-pass frequency filters, relative acoustic impedance and variance attributes from poststack seismic data. This approach supported stratigraphic analysis and helped in building a conceptual model of thin sand deposits at sub-seismic resolution. Flattening the seismic cube at the top of the Tertiary helped analyze reflection patterns within an incised valley. Natic topography influenced channelized sediment flow, creating favorable conditions of pathway for sand accommodation in fault-bounded spaces within a Tertiary rift setting. Despite limited well data, seismic indications of sand continuity were validated through limited electrical logs and mud logs.

Two wells were selected for recompletion; a high water cut producer, and an inactive water injector. Both intersected the shallow sand at geologically promising locations based on the seismic depositional model, and were perforated in the newly identified sand unit. They produced at good oil rates, confirming bypassed pay and validating the model. A performance difference between the wells aligned with predicted reservoir quality trends, supporting confidence in seismic interpretation.

The results supported planning the first horizontal well in this reservoir unit and prompted two more perforations to appraise sand in a different structural setting. This eliminated the need for costly appraisal wells, resulting in significant CAPEX savings and showcasing the value of integrating seismic interpretation with selective recompletion of underutilized wells. By targeting existing assets and applying a robust geological model, production was successfully enhanced from concept to execution. This work introduces a unique workflow combining advanced seismic interpretation with practical well screening using limited

mud log data correlated with washed-out logs. The novelty lies in identifying previously unrecognized stratigraphic targets and maturing reservoir potential cost-effectively without drilling new dedicated wells.

Introduction

The Early-Middle Paleocene (Al Hajri & Gomez, 2020) Shammar Formation is extensive throughout Oman Block-5 and covering wider areas in north Oman. The Formation is deposited directly above Base-Cenozoic/Cretaceous unconformity postdating immediately the first Alpine regional tectonic event in North Oman (Bazalgette et al., 2021). Shammar facies represent shallow marine deposit, primarily consisting of interbedded shale and limestone. It is also known as an effective seal in Block-5 where it lies unconformably on top of the Late Cretaceous Natih carbonate reservoir.

A discovery in Block-5 marked the first time the oil bearing Shammar sandstone being tested as prolific reservoir, exhibiting high porosity and permeability values typically ranging from 25 – 35% and 500 – 1000 mD, respectively. Generally, the total thickness of the Shammar Formation ranges from approximately 30 to 38 meters across Block-5, reaching up to 48 meters in certain areas. This extra gain in thickness is believed to be an indicator of the siliciclastic presence at the basal part of Shammar. This is believed to be due to the nature of the Base Tertiary unconformity where it forms localized topographic lows which accommodates the clasts and hence having a thicker Shammar which eventually traps hydrocarbons. The hydrocarbon is believed to be structurally and stratigraphically trapped within the reservoir, mainly in these localized topographic lows, sealed by shales of the overlying Shammar and underlying Fiqa or base Natih shale/tight units, sub-cropping under the base Tertiary unconformity (Fig.1).

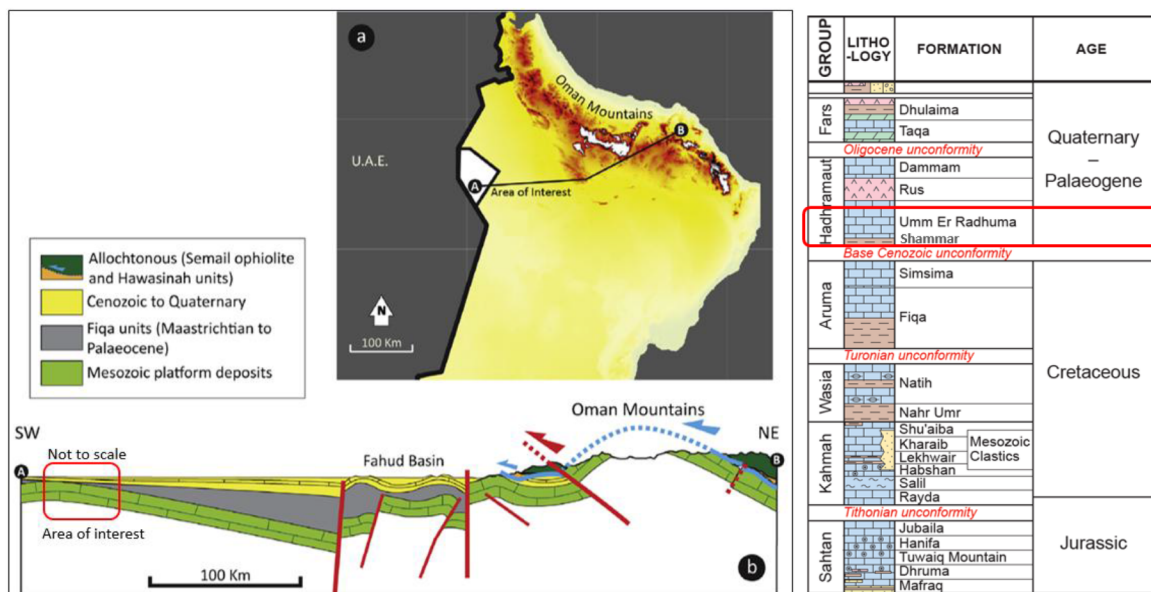


Figure 1—(Left) a simplified regional NE-SW cross-section through Oman Mountains, and the area of interest (modified after Bazalgette et al. 2018). (a) North Oman geographical map with cross-section location. (b) NE-SW cross-section. (Right) part of Oman lithostratigraphy (Forbes et al. 2010).

The siliciclastic Shammar member exhibits significant heterogeneity in terms of lithology and thickness. It consists of a thin, sub-seismic succession characterized by highly variable lithofacies, ranging from conglomerates to very fine-grained sandstones and siltstones. Thicknesses generally vary from approximately 1 to 5 meters. Notably, the presence of accommodation space does not necessarily correlate with reservoir quality, as it can be infilled with non-reservoir materials such as shale or extremely tight sandstones and siltstones.

Micro-image logs acquired from vertical wells showed no evidence of faulting or fracturing within the Shammar Formation. The main features observed were bedding planes with variable dip angles and a variety of sedimentary structures. These features included cross-bedding, high-angle bedding, thick bed sets, and erosional surfaces in certain intervals. In contrast, other sections displayed thin, flat bedding with heterolithic facies, varying degrees of bioturbation, and sharp contacts, highlighting the lateral and vertical heterogeneity within the formation.

Routine core analysis, supported by detailed sedimentological descriptions, indicated that the Shammar Sandstone succession predominantly comprises fine to very fine-grained stacked sandstone beds. These beds are often interbedded with siltstone laminae and beds. The entire succession reveals a clear upward-fining/muddying trend, accompanied by an increase in both the intensity of bioturbation and frequency of intercalated siltstone laminae and beds.

By integrating these observations along with additional well log data and seismic analyses, the evidence strongly suggests a tidally influenced depositional environment. This interpretation places the Shammar sandstones within the distal reaches of a fluvial channel system, likely situated near a paleo-shoreline or within an estuarine setting. In such an environment, tidal forces are sufficiently strong to alter and overprint the typical depositional patterns associated with river-dominated systems, resulting in complex sedimentary structures and facies variability that reflect both fluvial and tidal influences.

The cross-discipline integration of the data supported the development of a robust conceptual model. This integrated approach played an important role in guiding the successful appraisal & development journey. The same methodology was effectively applied across different areas of Block-5, ultimately contributing to a significant expansion of new oil development from this prolific reservoir unit.

Targeting the thin sandstone reservoir with a maximum thickness of approximately 5 meters, presented a significant challenge due to its thickness falling below seismic resolution threshold. To address this limitation, a comprehensive and densely interpreted seismic workflow was adopted, aimed at improving subsurface understanding and unlocking previously unrecognized reservoir.

The ability of seismic data to resolve subsurface features depends on the seismic wavelength, which is determined by dividing the velocity of the wave by its frequency. In seismic interpretation, a widely accepted rule of thumb is that vertical resolution (i.e. the smallest thickness of a layer that can be distinctly imaged) is approximately one-quarter of the wavelength. For the shallow Shammar reservoir, where the seismic wavelength is estimated to be around 25 meters, this means that features thinner than 6.25 meters fall below the resolution threshold. Consequently, such thin layers cannot be distinguished as separate reflectors on conventional seismic data, as illustrated in (Fig.2).

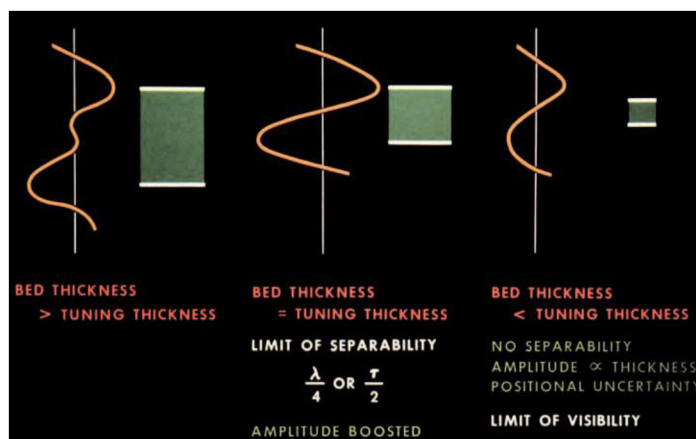


Figure 2—Resolution of reflections from a bed's top and base is controlled by the interference between closely spaced seismic wavelets (Brown, AR, 2004).

Multiple seismic attributes were integrated with available well data to infer the distribution and continuity of sand deposition which reach the maximum thickness approximately 3 meters in the targeted area. Although the reservoir could not be directly imaged due to resolution limits, these attributes enabled a preliminary delineation of the field's extent and geometry. This interpretation was further refined using advanced seismic analysis techniques, contributing to the development of a conceptual model for thin sandstone deposits at sub-seismic resolution.

Methods

The project primarily utilized post-stack seismic data, which had a frequency spectrum ranging from 10 to 60 Hz. Several processing steps were applied to the data, including multiple attenuation, Kx–Ky footprint removal, and bandwidth extension (BWXT). 3D seismic data volumes often display amplitude stripping patterns related to acquisition geometry, commonly referred to as the ‘acquisition footprint.’ This artifact results from variations in azimuth and offset that influence adjacent common depth points (CDPs). The Kx-Ky footprint removal technique mitigates this issue by applying a 3D frequency-wavenumber (FKxKy) filter to suppress both noise and acquisition-related artifacts in the stacked seismic volume. Bandwidth Extension (BWXT) is a technique used to broaden the seismic bandwidth at both the low and high ends of the frequency spectrum (Mahmoud and Yassein, 2018). This enhancement improves the resolution of the final stacked data, enabling more accurate interpretation and clearer delineation of reservoir features.

Well-to-seismic tie was used to identify the exact base Tertiary unconformity loop in well located near the area of interest. This well has its own vertical seismic profile (VSP) using the extracted rotated zero-phase statistical wavelet (Fig.3). Horizon interpretation was initially performed using 2D auto-tracking, followed by manual refinement in areas impacted by mispicks or low-confidence zones.

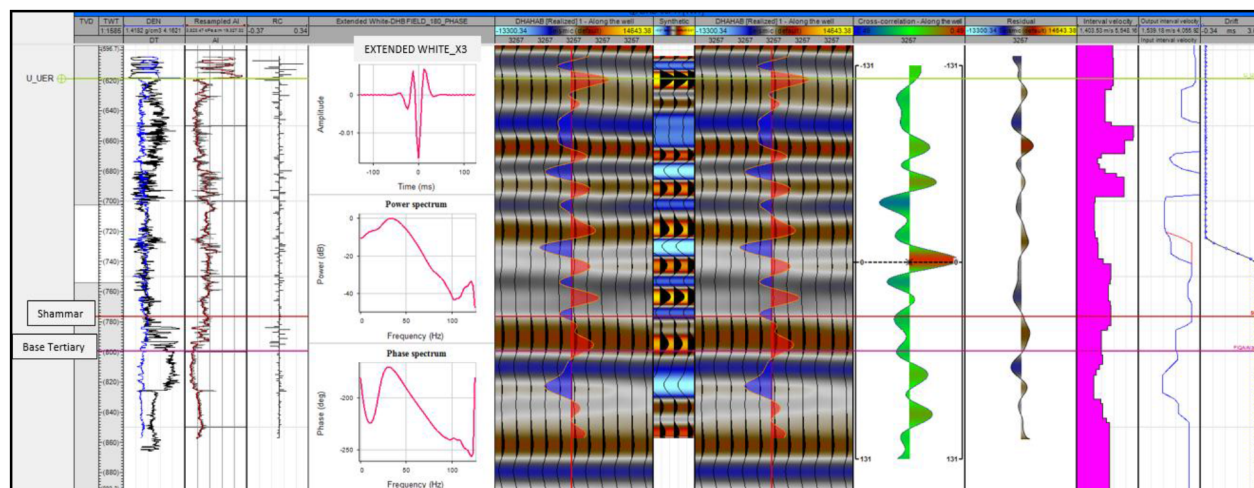


Figure 3—Seismic to well tie in X3 well using the extracted rotated zero-phase statistical wavelet.

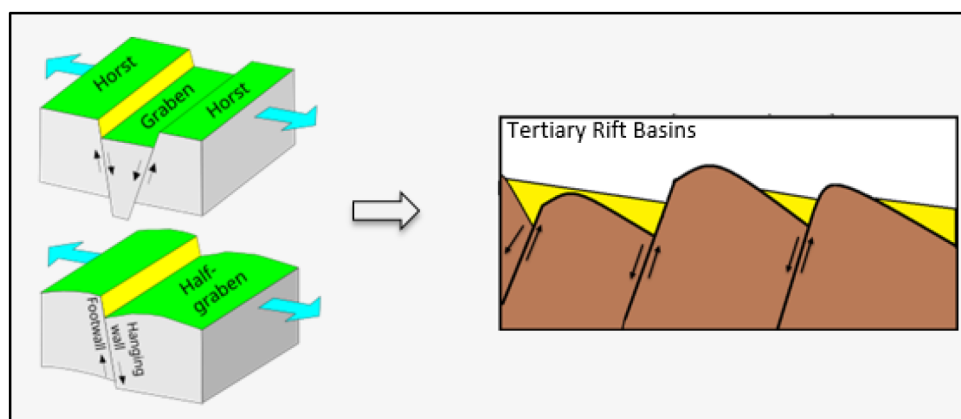


Figure 4—fault geometries forming the sand accommodation space by grabens or half-grabens.

Several seismic volume attributes were subsequently generated including: frequency filter, relative acoustic impedance and second derivative to reveal structural discontinuities and enhance reservoir delineation. Frequency filter was applied to filter out low frequency below 9 Hz which enhanced the seismic resolution in the Tertiary section and improved the clarity of surface interpretations. The relative acoustic impedance attribute was utilized to better define stratigraphic layering and distinguish subtle lithological boundaries. By integrating the seismic trace with low-cut filter to eliminate arbitrary long wavelengths, the sequence boundary was effectively enhanced in this field. In addition, second derivative attribute was used to guide the picking of the poorly resolved Top Shammar surface and fine-tune the interpretation of the unconformity surface (base Shammar) to resolve the subtle stratigraphic patterns. All the previous attributes were integrated to enhance horizon interpretation and reduce the ambiguity in picking the base of the Shammar Formation (Fig. 5).

Natih topography played a key role in directing channelized sediment flow, creating favorable conditions for sand accommodation, primarily present within fault-bounded accommodation spaces in a Tertiary rift setting within meanders of channel flow. These major faults creating grabens or half-grabens, which varied in size and acted as localized depositional basins. These basins served as accommodation zones for the deposition of both sand and shale over time (Fig. 4). Fine-grained shales or mudstones typically accumulated above these zones, acting as effective seal rocks for the localized, scattered basin traps. Seismic sections oriented northeast–southwest reveal Cretaceous-age fault systems that continue to influence the overlying Tertiary stratigraphy. Major fault trends were delineated, contributing to structural compartmentalization of the reservoir and potentially influencing hydrocarbon migration pathways (Fig. 5).

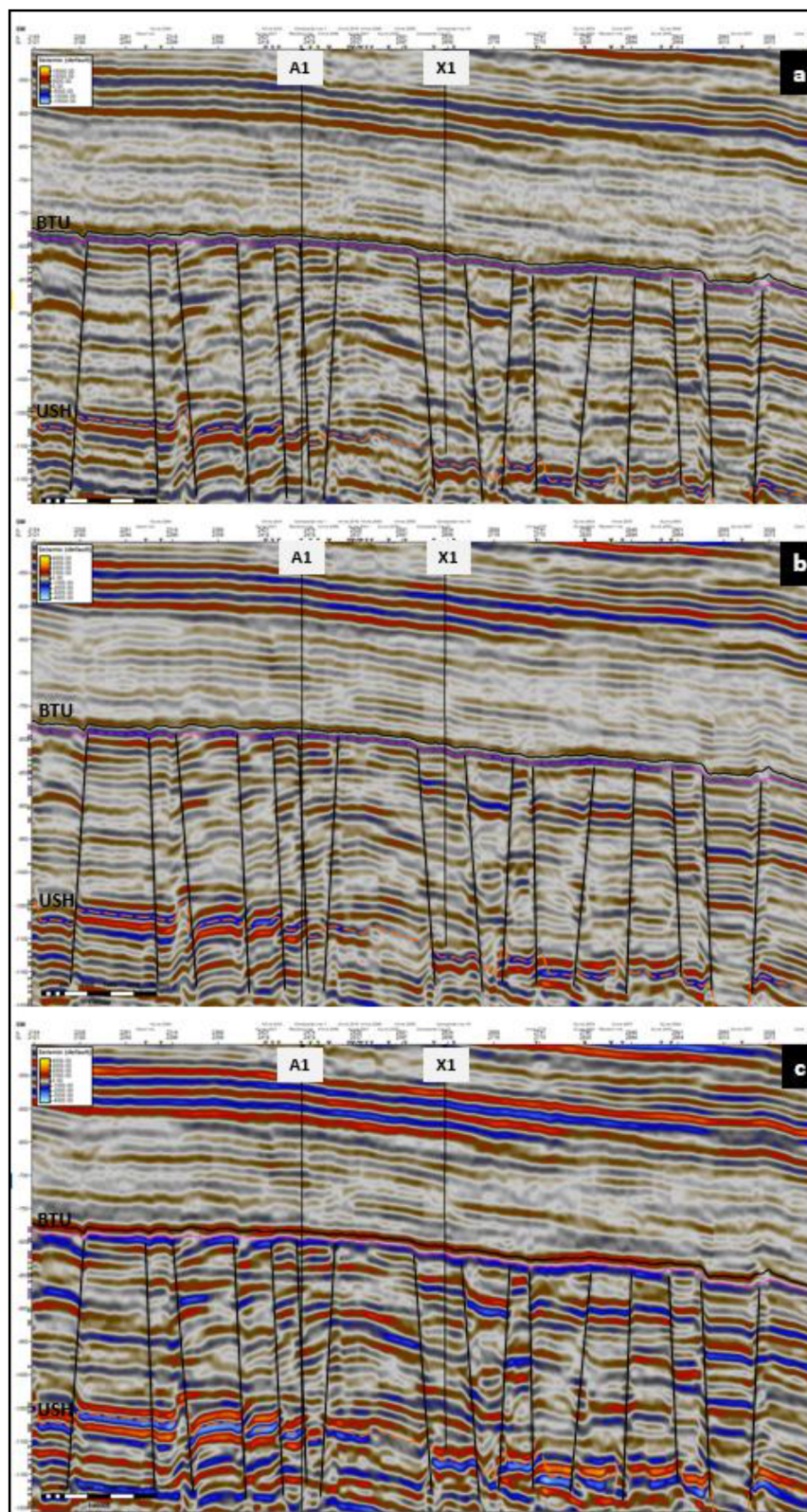


Figure 5—(NE-SW) Seismic enhancement; (a) Post-stack seismic data; (b) Second derivative; (c) Relative acoustic impedance.

Targeting a new reservoir which is below seismic resolution, in an area with limited well data presents significant geological and operational challenges. The availability of limited well data, such as gamma ray logs and mud logs in a limited number of wells, limits the geoscientist's ability to correlate lithological units

across the area and determine the continuity & extension of the oil-bearing sand. Fig.6 shows a correlation panel flattened along the Top Shammar. Although the wells are spaced only a few hundred meters apart, but a clear heterogeneity is observed in both the thickness & quality of the reservoir, making it very challenging to trace the reservoir unit laterally. Additionally, the reservoir interval in some of the wells is impacted by washouts leading to unreliable well log data such as well A3 in (Fig.6). Although the available well data was limited, the presence and continuity of sand indicated by seismic interpretation were validated through a comprehensive integration of this sparse well data with a robust geological conceptual model.

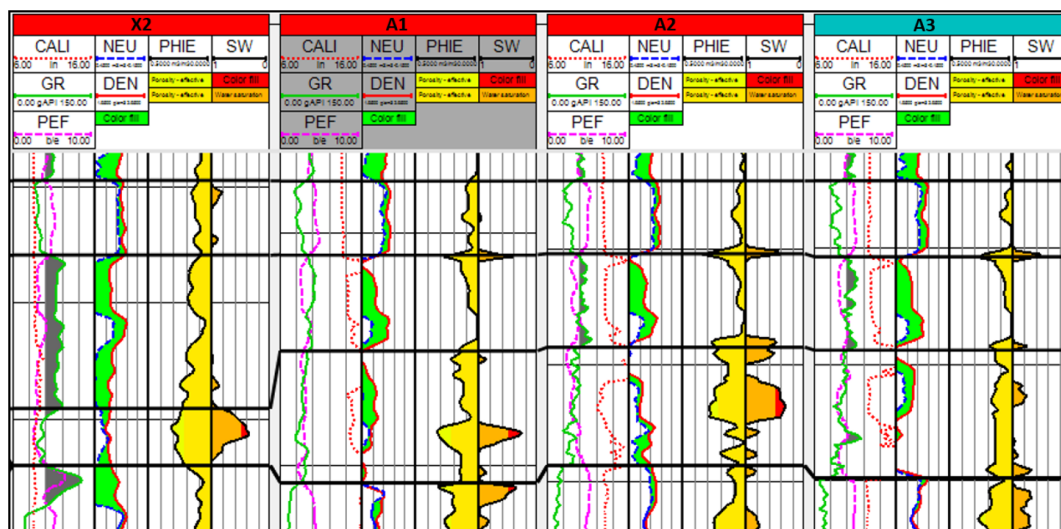


Figure 6—Well correlation for the available well data.

The seismic volume was flattened along the Top Shammar horizon to approximate paleogeographic conditions and reconstruct the structural lows at the time of deposition. Subtle structural lows remained evident after the flattening, representing small basin features that likely acted as localized sand accommodation zones. Furthermore, variance attribute analysis extracted on the Base Tertiary surface revealed a sinuous, meandering pattern, spatially aligned with the regional east to west sand thickness trends observed in Block-5 (Fig. 7). This alignment reinforces confidence in the seismic interpretation and provides further validation for the proposed depositional model.

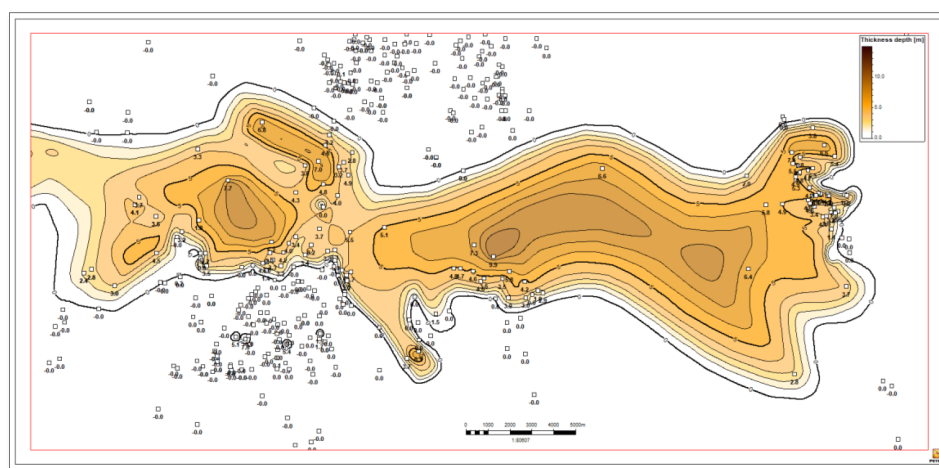


Figure 7—Shammar sand gross thickness map in block-5.

A densely interpreted time map of the Base Tertiary surface, covering the broader extent of Block-5, revealed several key structural features. These include subtle closures and a series of localized basins that exhibit progressive thickening of the sand interval toward the central part of the block. This thickening trend reflects a regional structural dip and suggests increased accommodation space in the basin (Fig. 8). These observations are in strong agreement with the sand thickness distribution presented in (Fig. 7), reinforcing the consistency between seismic interpretation and stratigraphic trends. Furthermore, a high-resolution structural map of the Base Shammar surface demonstrated a clear spatial correlation between thin sand accumulations and fault-bounded accommodation zones. This correlation was instrumental in identifying previously overlooked hydrocarbon-bearing intervals and refining the depositional model across the area.

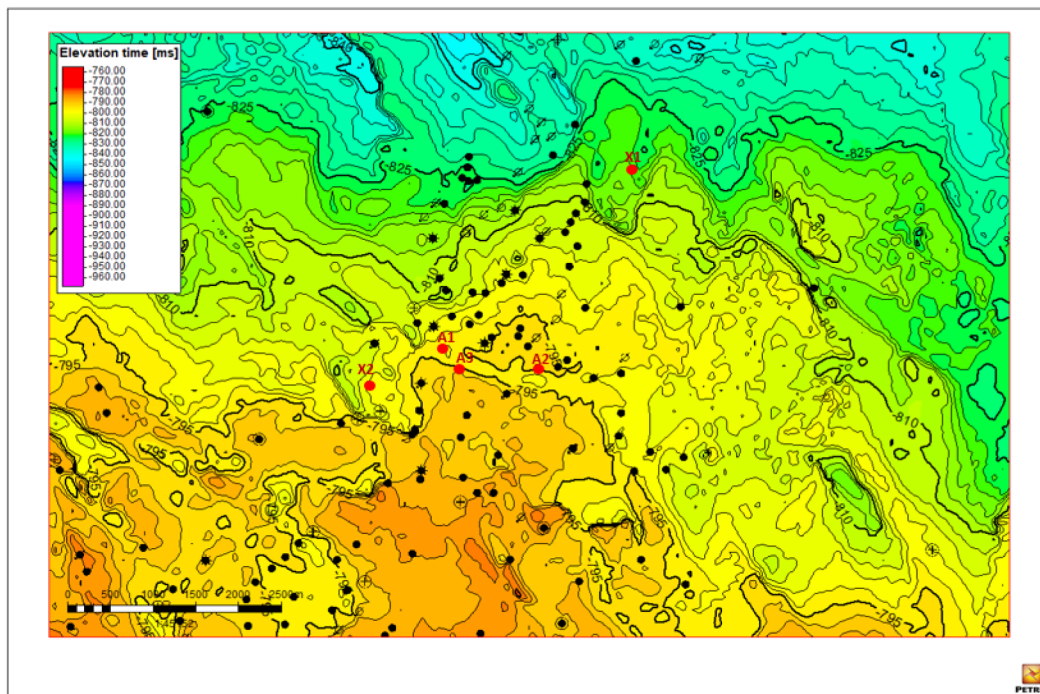


Figure 8—Base Tertiary time map after fine tuning the interpretation and building the concept

Recompletion

Two wells were ultimately selected as optimal candidates for recompletion: Well A2, which was producing with a high water cut, and Well A1, an inactive water injector. Both wells intersected the shallow sand in a geologically promising location based on the seismic depositional model and they showed encouraging indications from the available well data.

For both wells, the team carefully selected the cleanest and most geologically favorable intervals for perforation. To ensure effective perforation, an underbalanced perforation technique was employed (Fig. 9), allowing natural reservoir pressure to expel debris from the perforation tunnels and promote cleaner, and more uniform tunnel geometry. Furthermore, the perforated zones were treated with a 15% hydrochloric acid (HCl) solution for approximately 30 minutes.

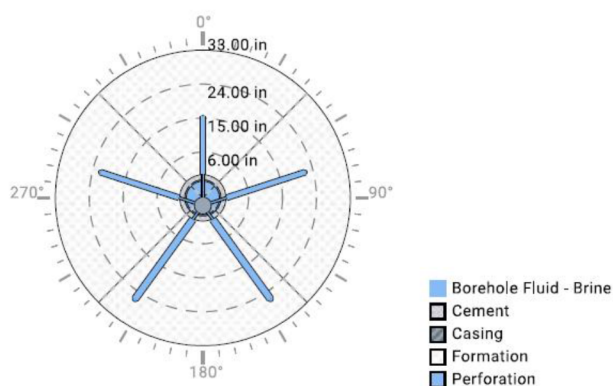


Figure 9—Penetration of the pure gun (underbalance) (SLB technical proposal for Daleel)

The primary objective of this acid treatment was to dissolve drilling-induced damage, including mud solids, crushed rock particles, and carbonate debris left behind after the perforation process. By effectively removing these obstructions, the acid treatment helped reduce near-wellbore skin damage. This, in turn, expanded the clean, open flow area around the perforation tunnels. As a result, both initial production rates and long-term flow performance were significantly improved (Fig. 10).

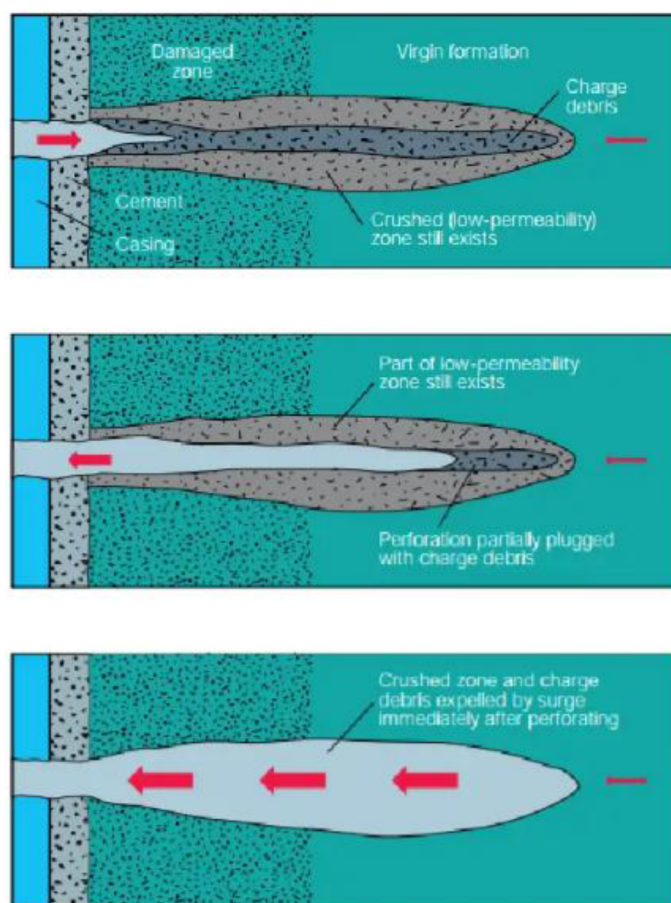


Figure 10—Dynamic Underbalanced Perforating (RYAN Petro Equipment)

Both A1 and A2 wells were completed using beam pumps and successfully produced at commercial oil rates, confirming the presence and productivity of the bypassed pay zone. This success validated the perforation strategy, demonstrated the effectiveness of the treatment method, and supported the seismic and

depositional model. Additionally, a notable performance disparity was observed between the two wells, due to various factors such as:

1. Reservoir quality trend

The production aligns with the reservoir trend, which improves toward the northwest of the targeted area. Well A1 shows better production performance than Well A2, reflecting the better reservoir quality in that direction.

2. Beam pump depth

(above or below the perforation, refer to the below graph (Fig. 11))

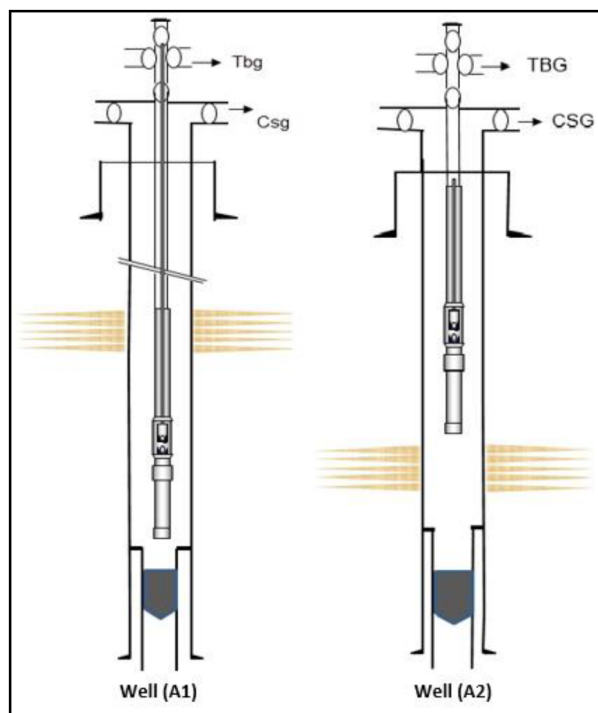


Figure 11—Beam Pump completion set below and above perforation for both wells

Results & Conclusions

The recompletion of two existing wells in the shallow sand reservoir yielded strong oil production rates, reinforcing earlier seismic and geological interpretations. This success not only confirmed the viability of the target interval but also served as a key stepping stone toward broader reservoir development planning.

The performance contrast between the two wells mirrored the spatial variability in reservoir quality, with higher productivity observed in zones previously predicted to have better reservoir characteristics. This outcome not only reinforced confidence in the seismic-derived depositional model but also highlighted its predictive value for future well planning.

The positive outcomes from the initial well recompletions directly supported the decision to drill the first horizontal well in this reservoir unit. Additionally, two more wells were selected for perforation to evaluate sand distribution in a different structural context. This phased, low-cost approach eliminated the need for new appraisal wells, resulting in an estimated 70% reduction in CAPEX compared to conventional drilling.

In addition, the strategy also mitigated the geological and operational risks typically associated with new horizontal wells by confirming productivity through existing wells. It also demonstrated the value of integrating seismic interpretation with selective recompletion of underutilized wells. By leveraging existing assets and applying a robust geological model, the team successfully enhanced production in mature fields from concept to execution.

This work introduces a unique workflow that combines advanced seismic interpretation with practical well screening, using limited well data and correlating it with challenging washed-out well logs in the targeted interval. The novelty lies in leveraging underutilized wells and advanced seismic interpretation to identify previously unrecognized stratigraphic targets, achieving cost-efficient reservoir development without the need for drilling new dedicated wells at the initial stage. The result from both two wells played a key role in shaping the plan for the upcoming horizontal appraisal well.

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